

Team 4 Requirements Document

- Mini World Description

- The mini-world that will be represented by this database is a zoo and its daily operations. The zoo contains animals, enclosures, employees, attractions, customers, ticket sales, food services, and a gift shop. The database will store and manage information about these areas so that the zoo can keep track of its operations and generate reports about its performance.

The database should allow the zoo application to:

- Maintain information about animals and their species.
- Track where animals are housed.
- Maintain information about zoo employees and their roles.
- Manage zoo attractions and the employees responsible for them.
- Store customer information.
- Record customer ticket sales.
- Manage products sold in the gift shop.
- Record gift-shop purchases.
- Manage food items sold by the zoo.
- Record food sales.
- Track revenue generated by different parts of the zoo.

Rules and Relationships

- Each Animal belongs to one species, while a species may represent many animals
- Each Animal is assigned to one enclosure, while one enclosure can have many animals
- Each employee belongs to one department, while each department may have many employees
- An enclosure can have many keepers and many veterinarians
- A customer could purchase multiple tickets
- A ticket purchase may have multiple ticket types
- A sale may have multiple products
- A supplier may have multiple products
- A product might have multiple suppliers

The system will be implemented using MySQL as the database, C# for the application logic, and Razor Pages for the web interface.

- Gift shop
 - GiftShop_ID
 - GiftShop_Name
 - Category
 - Location
 - Opening_Time
 - Closing_Time
- Product
 - Product_ID
 - Product_Name
 - Description
 - Category
 - Price
- Food stand
 - Stand_ID
 - Stand_Name
 - Description
 - Location
 - Category
 - Opening_Time
 - Closing_Time
- Food Item
 - FoodItem_ID
 - FoodItem_Name
 - Description
 - Price
 - Category
- Food_Sale
 - Sale_ID
 - FoodItem_ID
 - Quantity
 - Unit_Price
- Stand_Menu
 - Stand_ID
 - FoodItem_ID
 - Price
- Inventory
 - Product_ID
 - Inventory_ID
 - GiftShop_ID

- Quantity_In_Stock
 - Reorder_Level
 - Last_Updated
- Supplier
 - Supplier_ID
 - Supplier_Name
 - Phone
 - Email
 - Address
 - Delivery_Schedule
- Supplier_Product
 - Supplier_ID
 - Product_ID
 - Unit_Cost
 - Availability
 - Delivery_Schedule
- Sale
 - Sale_ID
 - Sale_Date
 - Stand_ID
 - Employee_ID
 - Customer_ID
 - GiftShop_ID
 - Total_Amount
 - Payment_Method
- Merchandise_Sale
 - Sale_ID
 - Product_ID
 - Quantity
 - Unit_Price
- Customer
 - Customer_ID
 - First_Name
 - Last_Name
 - Email
 - Phone Number
- Ticket_Type
 - TicketType_ID
 - TicketType_Name
 - Price

- Eligibility_Requirements
 - Children 6 and under (free)
 - Student
 - Adult
 - 65_and_up
 - Veteran
- Memberships
 - Membership_ID
 - Customer_ID
 - MembershipType_ID
 - Start_Date
 - Expiration_Date
 - Status
- Membership_Type
 - MemberShipType_ID
 - Membership_Name
 - Price
 - Benefits
 - Tier_1(Standard)
 - Tier_2(Mid-Tier)
 - Tier_3(Premium)
- Ticket_Sales
 - TicketPurchase_ID
 - Customer_ID
 - Purchase_Date
 - Total_Amount
 - Payment_Method
- Ticket_Purchase_Item
 - TicketPurchase_ID
 - TicketType_ID
 - Quantity
 - Unit_Price
- Employee
 - Employee_ID
 - SSN
 - First_Name
 - Middle_Name
 - Last_Name
 - Address (1234 House Ave. Houston, TX, 11022, United States)
 - Gender

- Phone_Number
 - Job_Position
 - Hire_Date
 - Department_ID
 - Employment_Type
 - Pay_Type
 - Pay_Rate
- Job_Position
 - Position_ID
 - Position_Name
 - Description
- Shift
 - Shift_ID
 - Employee_ID
 - Shift_Date
 - Start_Time
 - End_Time
- Veterinarian
 - Employee_ID
 - Specialization
- Keeper
 - Employee_ID
 - Experience_Level
- Keeper_Enclosure
 - Employee_ID
 - Enclosure_ID
 - Assignment_Start_Date
 - Assignment_End_Date
- Volunteer
 - Volunteer_ID
 - First_Name
 - Last_Name
 - Phone_Number
 - Email
 - Start_Date
 - End_Date
 - Volunteer_Status
 - Department_ID
- Attraction
 - Attraction_ID

- Attraction_Name
 - Attraction_Type
 - Location
 - Price
 - Status
- Employee_Attraction
 - Employee_ID
 - Attraction_ID
 - Assignment_Date
 - Role
- Animal
 - Animal_ID
 - Animal_Name
 - Species_ID
 - Gender
 - Date_of_Birth
 - Health_Status
 - Enclosure_ID
 - Date_Acquired
 - Acquisition_Type
- Species
 - Species_ID
 - Category_ID
 - Common_Name
 - Scientific_Name
 - Diet_Type
 - Conservation_Status
 - Habitat_Type
- Enclosure
 - Enclosure_ID
 - Enclosure_Name
 - Enclosure_Type
 - Location
 - Capacity
- Feeding
 - Feeding_ID
 - Animal_ID
 - Employee_ID
 - Feed_ID
 - Feeding_Date

Isabella

- Feeding_Time
 - Quantity
 - Feeding_Notes
- Feed
 - Feed_ID
 - Feed_Name
 - Feed_Type
 - Supplier_ID
 - Quantity_In_Stock
 - Storage_Location
 - Expiration_Date
- Animal category
 - Category_ID
 - Description
 - Category_Name
 - Mammal
 - Bird
 - Reptile
 - Amphibian
 - Fish
 - Invertebrate
- Department
 - Department_ID
 - Department_Name
 - Location
 - Phone_Number
- Maintenance Sazid Afridi
 - Maintenance_ID
 - Description
 - Location
 - Request_Date
 - Completion_Date
 - Status
 - Cost
 - Employee_ID
 - Enclosure_ID
 - Attraction_ID
 - Stand_ID
 - GiftShop_ID

Zoo Database Schema

